

Jayakodige Lasith Charuka Perera

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Professional Summary

Motivated Computer Science undergraduate majoring in Artificial Intelligence at Swinburne University of Technology with practical experience in machine learning, software engineering, database systems, and full-stack application development. Skilled in Python, Java, TensorFlow, SQL, and data analytics with experience developing AI-based predictive systems and web applications. Passionate about building intelligent and scalable software solutions to solve real-world problems.

Technical Skills

- **Programming Languages:** Python, Java, JavaScript, PHP, C++
- **Machine Learning & AI:** TensorFlow, Scikit-learn, XGBoost, LSTM, GRU, Data Preprocessing, Feature Engineering
- **Web Development:** HTML, CSS, Streamlit, Flask
- **Databases & Tools:** MySQL, Git, GitHub, VS Code, Jupyter Notebook, Pandas, NumPy, Matplotlib

Projects

Traffic Flow Prediction System

- Developed machine learning and deep learning models including LSTM, GRU, CNN, and XGBoost for traffic flow prediction using SCATS datasets.
- Applied feature engineering and preprocessing techniques to improve prediction accuracy.
- Integrated predictive models with evaluation metrics and visualizations.
- Implemented route-guidance and travel-time estimation logic.

Archery Score Prediction System

- Designed and implemented a relational database system for archery score management using MySQL.
- Developed an XGBoost-based prediction model using historical performance data.
- Built a Streamlit application for CSV uploads and real-time predictions.
- Implemented data preprocessing and visualization for model transparency.

Olympic Medal Prediction Model

- Built regression-based machine learning models to predict Olympic medal counts using historical datasets.
- Improved model performance through feature engineering and optimization techniques.
- Reduced prediction error using Ridge Regression and evaluation tuning.

Education

Bachelor of Computer Science (Artificial Intelligence)

Swinburne University of Technology

Expected Graduation: August 2026

Certifications

- Generative AI: Working with Large Language Models
- Introduction to Large Language Models
- Introduction to Generative AI
- Introduction to Responsible AI
- Machine Learning Fundamentals

Additional Information

- Strong interest in Artificial Intelligence and Software Engineering
- Active GitHub portfolio with machine learning and software development projects
- Strong problem-solving and analytical skills